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ChatGPT and university teaching, learning and assessment: some initial reflections on teaching academic integrity in the age of Large Language Models

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Since its arrival in late 2022, ChatGPT has occupied the minds of academics, administrators and students. Reactions to the emergence of Large Language Models (LLMs) have varied but significant anxieties about their impact on assessment have arisen. To address these concerns, this article serves three purposes; firstly, it seeks to gauge the discourse surrounding Large Language Models (LLMs) focusing on ChatGPT. In doing so, it explores general and academic responses to the technology and the challenges/opportunities that have been identified. Secondly, and building on this, it provides an overview of ChatGPT in action and seeks to moderate fears raised by some of the extreme claims that have been made about the potential of this technology. Finally, the article uses a case study to discuss the introduction of this technology to a group of first-year undergraduate students, offering guidance on how the topic of LLMs might be broached. It concludes by suggesting that while the technology has the ability to offer assistance in the completion of academic assessment, it does not replace the higher thinking skills that are central to teaching, learning and assessment in higher education. In turn, arguing that these must be central to assessment practices.

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Introduction

ChatGPT (one of a number of Large Language Models (LLMs)) burst into our lives like a new-born: all-consuming, met by equal amounts of worry and excitement and with little comprehension as to what it will be when it matures. Academics and educational administrators have scrambled to understand and react to the perceived dangers that it represents to the integrity of assessment. While some, such as the New York Education Department, have blocked access to the technology, others seek to integrate it into learning and assessment (Ceres 2023; Elsen-Rooney 2023; Wallbank 2023; Weale 2023). Notwithstanding the newness and ingenuity represented by ChatGPT, public reaction to it has followed well-worn tropes around emerging (educational) technologies (see Blickstein and Blickstein 2021). Suggested positive uses have included possible AI assisted

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instruction, including personalised feedback and learning. While others have discussed its role in the demise of the academic essay – to the horror or delight of educators depending on their attitude to that much maligned form of assessment – and the need to impose in-person exams and viva-voce to ensure academic integrity (Cassidy 2023; Mollick and Mollick 2023; Stokel-Walker 2023b; Zhai 2023).

Despite the furore surrounding LLMs, the smorgasbord of conflicting responses to it demonstrates limited understanding of the technology and its potential to ‘disrupt’ higher education. This article argues that, contrary to suggestions that we ban or ignore these technologies, we should rather involve students as partners in their exploration. This will allow for greater critical engagement with generative AI, in turn enhancing academic integrity. For educators, this means exploring the affordances and limitations and in turn the potential uses and misuses of LLMs. Equipped with this knowledge, they can do what they have always done: prepare students to ask deep questions about the factors that shape their world and how they will interact with them. By reflecting on such efforts with a group of first-year students at Maynooth University, this article will provide an understanding of not only how this technology functions, including its strengths and weakness, but also seeks to provide insights on how to best encourage students to think critically about their own potential use of AI. To do this, it will first explore current reactions to ChatGPT; then discuss the technology’s operation, affordances and limitations; and finally add some reflections on introducing undergraduate students to the technology and the role that this can play in enhancing academic integrity.

Literature review

The arrival of ChatGPT has generated much popular and scholarly interest in the potential of ‘AI’ to ‘disrupt’ higher education. Reports of academic papers written in part by ChatGPT and other generative AI models (including GPT-3, a precursor to ChatGPT)¹ have no doubt added to fears of the final demise of the academic essay. However, editorial guidelines have been quickly updated to disavow such authorship claims, in a large part because LLMs cannot take legal and reputational responsibility for the material published (King 2023; Stokel-Walker 2023). Despite this, it appears that authors are still going to use ChatGPT. For example, in April (2023) the American Psychological Association issued guidelines on citing text produced using ChatGPT (APA 2020).

A commentary published in *Nature* on 3 February 2023 gives some insight into the uses to which the technology has been put: ‘researchers and others have already used ChatGPT and other LLMs to write essays and talks, summarise literature, draft and improve papers, as well as identify research gaps and write computer code, including statistical analyses’. Nevertheless, in claiming that ‘soon this technology will evolve to the point that it can design experiments, write and complete manuscripts, conduct peer review and support editorial decisions to accept or reject manuscripts’, the article also provides insights into the exaggerated claims made about the future of the technology (van Dis et al. 2023).

There can be no discounting the paradigm shift which ChatGPT has brought about. Recent research has highlighted its ability to pass ‘graduate-level exams in different disciplines, including law, medicine, and business’ (Rudolph, Tan, and Tan 2023) and its

potential uses in medicine (Ayers et al. 2023; Johnson et al. 2023). A lot of attention has been drawn to a working paper published on the preprint website Arxiv. Of the four authors one is affiliated with OpenAI (the company behind ChatGPT), and one with OpenResearch (a company involved in the production of software solutions), one with both OpenResearch and OpenAI and the final one with the University of Pennsylvania (Eloundou et al. 2023). In this the authors, using a rubric that they have created for this article, claim that 80% of the U.S. workforce could have at least 10% of their tasks impacted by LLMs, while 15% of the workforce could see at least 50% of their tasks affected. Despite the rather dramatic statements as to the inevitability of LLM's playing an increasing role in the workplace of the future, we must be careful of the boosterism that surrounds this discourse. Even these authors' predictions of the future are qualified with an understanding that 'social, economic, regulatory, and other determinants imply that technical feasibility does not guarantee labor [*sic*] productivity or automation outcomes' (Eloundou et al. 2023).

Rudolph, Tan, and Tan (2023b) identify a number of potential opportunities which ChatGPT raises, seeing it as an opportunity to rethink assessment and utilise this for, as and of learning. It suggests use of flipped classrooms (with assessment focused in class), multimedia assignments and oral presentations as alternative assessment practice. They also highlight various options forwarded to overcome the use of generative text, including testing questions against AI, analyse of images, videos, class discussions, or texts that are too long for ChatGPT to analyse. They note that currently ChatGPT cannot insert intext references (although various plugins are appearing that seek to draw information from the web to address this).

Concerns about the technology centre on three central themes: the potential infringement of copyright, privacy concerns and, for educators, its impact on academic integrity. The first two can be summarised briefly, while the third for the purpose of this article requires some deeper consideration. Copyright and privacy concerns centre around the 'training' of these system. GPT-3.5, its progeny ChatGPT, and its latest incarnation GPT-4, are trained on large tracts of texts scraped from the internet. This presents similar problems to Generative Art AI which do not allow content creators to opt out of inclusion in training data (Burrell 2023). As such, content creators have considerable concerns about the use of their text in training, as it can then replicate similar text without acknowledging and recognising the copyright of the original authors.

The significant size of this training data and ambiguity regarding its content has raised privacy and data protection concerns. Italy, for example, banned ChatGPT citing unease about the legal basis surrounding the use of personal data in its training data (a decision it subsequently revoked) (Milmo 2023). Elsewhere politicians have voiced similar unease surrounding the lack of AI regulation (Volpicelli 2023). This has been exacerbated by OpenAI's (the company behind ChatGPT) coyness surround its training data (Burgess 2023; Chang et al. 2023; Fraser 2023; Hern and Milmo 2023; Mukherjee, Pollina, and More 2023). In addition, conversations with ChatGPT are used for further training of the model, although an opt-out from this was announced in April 2023 (Gold 2023). While these general concerns are of interest to many, for educators the potential impact of these technologies on academic integrity has formed a central discursive theme.

The perceived threat of ChatGPT to the integrity of academic assessment is a problem that has grasped the most attention within higher education. Cotton, Cotton, and Reuben Shipway's (2023) recent article on 'chatting and cheating' begins with a substantial ChatGPT generated text (this is kept hidden from the reader until mid-way through the article, a rhetorical device that has nearly become a genre when discussing AI)² to demonstrate the cohesion of such 'AI' generated text. While the text does flow and read as produced by a human author, the information and 'understanding' demonstrated is problematic. For Cotton, Cotton, and Reuben Shipway (2023), GPT-3.5 (this and GPT-4 are the LLMs upon which the chatbot ChatGPT is built) represents a number of challenges for academic integrity. They identify possible plagiarism as the most significant, as 'it can be difficult to distinguish between a student's own writing and the responses generated by a chatbot application'. The popular press has also raised the issue of academic integrity brought about by the arrival of ChatGPT. A recent article in the popular magazine *WIRED* connects the emergence of concerns surrounding LLMs and plagiarism to those that surround improper use of the internet, however 'the only difference now is that the internet can think for itself' (Barnett 2023). However, are such claims about the technology warranted? A brief look at how LLMs operate should help to temper such fears.

ChatGPT: how does it work

At its most basic level ChatGPT, as with other LLMs, seeks to produce plausible writing based on the text that has already been produced/supplied (the architecture for this LLM is GPT, Generative Pre-trained Transformer). There are two clear reasons for its current popularity; firstly, unlike the subscription service utilising GPT-4 it is free to use and, secondly, it is easy to use thanks to an application programming interface. This allows the technology to reply to prompts entered in natural language. For example, one simply enters a prompt into the prompt bar and it will nearly instantaneously produce coherent text in response. So if we enter the incomplete sentence 'AI can help you ...' and ask it to add one word ChatGPT will produce the following:

'AI can help you immensely'. (OpenAI 2023)

While this may seem like the ultimate text prediction software, it is much more than this and can continue independently writing significant amounts of text. GPT-3.5 can facilitate conversations with 4,096 tokens, while the subscription service GPT-4's limit is 32,768 tokens. This represents both prompts and replies. Tokens are pieces of words, 100 tokens are around the equivalent of 75 words (OpenAI n.d.). It is able to predict the most likely word to follow any given text (as the training corpus is of human writing this prediction thus renders quite human-like writing). Cleverly, it does not select the most probable word each time but rather one of the most likely. In addition, unlike basic text prediction that one would see on a digital device, ChatGPT does not just predict the probable next word but also the probability of sequences of words (Wolfram 2023).

To produce coherent text the technology is trained on human writing which leads to the production of a neural net of 175 billion parameters (connections) that are drawn on each time it calculates the next word (this is in comparison to the 1.5 billion in OpenAI's earlier LLM, GPT-2). Thus, it does not store its training data but develops a network of parameters which are used to generate statistically probable text. The final part of its

training involves interaction with humans. While the text it produces at this point is statistically correct, OpenAI employs human operators to fine-tune the LLM's output by providing feedback on its responses to prompts. In addition, OpenAI also inserted guardrails to address any potential harmful outputs (for example racist language) (OpenAI 2023). These processes combine to not only produce more human-like writing that is considerably longer than LLMs have hitherto produced, but also text that is different each time, even if the same prompt is entered (Wolfram 2023). Taking our earlier example, 'AI can help you ...', it produces these responses when prompted multiple times to add one word to the sentence:

AI can help you achieve your goals.

AI can help you tremendously.

AI can help you immensely. (OpenAI 2023)

This leads to two important and worrying issues for educators. Firstly, ChatGPT does not replicate any single piece of writing, rather it uses its vast network of parameters to predict the most likely text. Consequently, traditional detection software used to identify similarities between student work and other text is of little use as the text is not plagiarised from elsewhere. Secondly, each time a prompt is entered it will iterate different text and as such will not show similarity with other essays produced using the technology, again making traditional plagiarism detection software useless. It should be noted that there are ongoing efforts to produce AI detection software but at the moment while these claim high levels of accuracy they can only indicate possible use of AI (Gao, Howard, and Markov 2023; NAIN 2023; Turnitin n.d.). Despite these concerns, it is important to think further about the text produced. In response to the prompt 'Write a paragraph on the benefits of wind produced electricity' it produced the following (passable) text:

Wind energy is a clean and renewable source of electricity that has numerous benefits. One of the primary advantages of wind-produced electricity is its environmental friendliness. Unlike fossil fuels, wind power does not produce harmful emissions or pollutants that contribute to climate change and air pollution. In addition, wind energy is a domestic source of energy that helps to reduce our dependence on foreign oil, thus increasing energy security. Moreover, wind energy can be produced in both large-scale and small-scale operations, making it an accessible option for communities, businesses, and homeowners. Overall, wind-produced electricity is a sustainable and reliable source of energy that provides numerous benefits to the environment and society as a whole. (OpenAI 2023)

On initial analysis, the above text appears authoritative, coherent and factually correct. However, despite the recent advances in LLMs, one thing they cannot do is think. ChatGPT is merely producing statistically probable text based on the parameters that have been developed during its training. Consequently, it will often confidently present inaccurate information as fact, a phenomenon that has been termed 'hallucinations' (Alkaiissi and McFarlane 2023). This issue is inherent in LLMs, as the technology is not drawing on any real sources in its responses, rather is producing the statistically most likely text. Due to this, while upgrades might improve the fluency and length of text produced, it will not remove this propensity for misinformation. For example, when prompted to 'produce a reading list on technology in Irish history', none of the sources it presents exist:

Here's a list of books and articles that could be of interest for those wanting to learn about technology in Irish history:

Books:

- 'The Hidden History of Ireland's Technology' by Niall O'Brien (2018)
- 'A History of Irish Technology' by Philip O'Connor (2021)
- 'Irish Inventions and Discoveries' by Sean McCarthy (1997) (OpenAI [2023](#)).

It produces these non-existent texts for a simple reason, it does not have access to the original sources but rather using the parameters developed during its training is producing the statistically most likely wording of a source. Consequently, while LLMs are quite suitable for text that is *pro forma* (as seen in the wind energy reply above). It encounters many more difficulties when questions are asked of it that seek demonstration of higher thinking skills, such as analysis, synthesis, and evaluation and through these the production of new arguments and knowledge (Armstrong [2010](#)). For these reasons LLMs will often produce text that reads well but is nonsensical or lacks the nuance expected of academic writing that is balancing many interpretations of historical events, social issues, philosophical debates and so on. This raises a number of issues that are really essential for educators to engage students with, particularly given that these technologies are not going away (ChatGPT is currently free to use and Microsoft is in the process of integrating a LLM based on this technology into its Office programmes (Spataro [2023](#))).

Perhaps the key challenge presented by the development of these technologies is the most obvious: students are going to use them. Yet, unlike the academics discussed at the start of this article who have used these technologies to support their writing, students have not yet developed the skills in writing, research and critical thinking nor the subject-matter expertise to critique what LLMs produce. This points to a number of challenges; firstly, if students attempt to use these technologies early in their degree programmes there can be little doubt that this will hinder the development of writing, research and subject-matter knowledge. For example, studies of first-year seminars in the US (which share many elements of Maynooth's Critical Skills course) have been shown to increase student retention and performance (Krsmanovic [2019](#); Maxson [2022](#)). This raises significant concerns that the use of LLMs will not only hinder their development as scholars but also impact their grades as assessment becomes more difficult as they progress through their studies. Likewise, even if the use of these technologies is adopted at a later stage, lack of disciplinary knowledge can seriously impact students' grades, progression and employability. Thus, rather than concerns surrounding academic integrity, which are of course legitimate, for most students the real issue is the negative impact this may have on their educational development. Consequently, educators must equip students with an understanding of this technology so that they can think critically about its affordances and limitations.

Introducing ChatGPT to the classroom

The rest of this article will explore the introduction of ChatGPT to a group of first-year undergraduate students who were undertaking the Critical Skills programme at Maynooth University (2022-23). In doing so, it seeks to give a snap shot in time, demonstrating the abilities of ChatGPT as they existed when the class was conducted, while seeking

to demonstrate the types of questions we must ask of these technologies. Central to developing critical AI literacies is an understanding that these technologies are constantly evolving, as such one cannot make universal claims as to what they will produce in response to prompts. Generative AI is a powerful and impactful development which engineers hope to further improve and refine. However, it is also important to state that it has limitations that are inherent in its current architecture, i.e. a LLM is not an AGI (Artificial General Intelligence) (Bender et al. 2021; Goertzel 2023; Stahl and Eke 2024). These difficulties are perhaps best summed up by Sam Altman (CEO of OpenAI) in a tweet on 11 December 2022,

ChatGPT is incredibly limited, but good enough at some things to create a misleading impression of greatness. It's a mistake to be relying on it for anything important right now. it's [sic.] a preview of progress; we have lots of work to do on robustness and truthfulness.' (Altman 2022)

Nevertheless, despite current limitations, we must be cognisant that companies such as OpenAI are actively developing and updating these technologies.

Critical Skills is a credit-bearing, year-long subject choice for first-year students at Maynooth University. Each class group contains a maximum of 25 students drawn from across the humanities and social sciences. The course focuses on developing transferable academic skills, including research, writing, numeracy and oral presentation skills, as well as developing analytical and critical thinking. The first semester of the course focuses on research and writing skills. One of the key learning outcomes for students is the ability to write an academic essay and as such they develop a strong understanding of this genre of writing. To do this the students research and complete an academic essay on a theme agreed between the class and instructor (each student focusing on a particular topic within the class theme). Consequently, students have gained experience in academic writing prior to their introduction to ChatGPT early in semester two.

The introductory one-hour class sought to demonstrate how to operate ChatGPT but also to provide students with an opportunity to identify the affordances and limitations they face in using it and the potential negative impact of its use on their own learning. This aligns with Concannon et al.'s (2023) recent call to develop students' critical AI literacies. Drawing on Southworth et al. (2023), they identify a number of AI literacy types students will need: enabling AI; knowing and understanding AI; using and applying AI; evaluating and creating AI; AI ethics. Thus, critical AI literacies should allow 'students to ask critical questions about technology' (Concannon et al. 2023, 3). The lesson was conducted with two class groups (24 students in total) in February of 2023 (early in the second semester). The instructor and students made use of the free version of ChatGPT (based on GPT-3.5) throughout the class, as the subscription model had not been released. The intended learning outcomes were that students would be able to discuss the affordances and limitations of LLMs (focusing on ChatGPT for this class); utilise it to produce text; critique the text it produced; and, finally, appraise the potential uses of this technology in their own (scholarly) work.

The class firstly introduced the technology in a general sense, how it operated, as well as ethical, legal and privacy concerns (i.e. much of the discussion that is outlined above). The technology was then demonstrated by entering the prompt 'Write a business plan for an exploration of Mars'. Its response was:

The purpose of this business plan is to outline the feasibility of launching a Mars exploration mission. The objective of the mission is to study the geology, climate, and potential for human habitation on Mars. The ultimate goal is to establish a permanent human settlement on Mars, which would provide numerous benefits to both humanity and the advancement of scientific research.

Market Analysis:

The market for space exploration and the study of Mars is growing rapidly. With the increasing interest in space exploration and the development of new technologies, the demand for Mars exploration missions is expected to continue to rise. This demand is driven by both government agencies and private organizations, which are investing in space exploration to advance scientific research and to secure a future for humanity beyond Earth ... [see appendix 1 for complete text response]

Having used this example to demonstrate the technology, a further prompt was entered to demonstrate the iterative nature of the conversation (i.e. it retains knowledge of previous prompts and responses in the conversation and can utilise these in responding to further prompts), its ability to generate different text responses, and to write in different voices. It was prompted to 'now write this in the style of a pirate'. Its response was:

Ahoy, mateys! 'Tis time to set sail on a grand adventure to the red planet, Mars! We be embarkin' on a mission of exploration, with a goal of studyin' the geology, climate, and potential for pillagin' and colonizin' the new world.

The market be ripe for this journey, with many a government agency and private organization investin' in space exploration and the search for new treasures. Our offerings be vast, from scientific research to new technologies and even the establishin' of a permanent pirate settlement on Mars ... [see appendix 2 for complete text response]

While, of course, this provided some joviality, it also demonstrated a powerful element of the technology: its ability to work with previously generated text and modify this according to new prompts. Thus, having demonstrated the basic functions of the technology the class moved on to a more critical evaluation.

Central to the successful use of any technology, is a knowledge of its abilities. LLMs have many affordances but also limitations and without a proper understanding of these, students cannot begin to think about how to integrate any technology into their studies, writing and everyday life. To facilitate this discussion, the class focused on an exploration of the Irish famine (in a disciplinary specific class something that aligned with previous teaching and learning would be ideal). Two prompts were used for this, the first a general one 'Write a 500 word essay on the causes of the Irish famine', which produced text that was passable and gave a general overview of the causes of the Irish famine:

The Irish Famine, also known as the Great Hunger, was a catastrophic event in Irish history that took place between 1845 and 1852. It was a period of extreme poverty, suffering, and death caused by the failure of the potato crop, the main staple food of the Irish people. The famine had a profound impact on Ireland and its people, causing mass emigration and reducing the population of the country by approximately 25% ... [see appendix 3 for complete text response]

To follow up on this, a more complex prompt was entered in the conversation 'add a discussion of the role of land holding'. This in comparison to the first question is more

nuanced and requires the balancing of various interpretations (something much more difficult for ChatGPT). In response it produced the following text:

Another significant factor that contributed to the Irish Famine was the issue of land holding. The majority of the Irish population were tenants, renting their land from wealthy landlords, mostly of British origin. The system of land holding in Ireland was unjust and unfair, with tenants paying high rents for small plots of land that were often inadequate for their needs. This left the Irish people vulnerable to poverty and food insecurity, as they were unable to grow enough food to feed themselves and their families.

The issue of land holding was made worse by the practice of eviction, which was a common occurrence during the famine. Landlords would evict tenants who were unable to pay their rent, forcing them to abandon their homes and land. This not only left the tenants homeless, but also made it difficult for them to find food and support, as they had no means of growing crops or earning a livelihood.

Furthermore, the British government's response to the famine was heavily influenced by the interests of the wealthy landlords. Instead of providing direct aid to the Irish people, the government focused on policies that would benefit the landlords, such as reducing tariffs on imported food, which made it easier for landlords to import food to sell at high prices. The government also failed to enforce laws that would have protected the rights of tenants, such as the Protection of Tenants Act of 1847, which would have prevented landlords from evicting tenants during the famine.

In conclusion, the issue of land holding played a significant role in the Irish Famine. The unjust and unfair system of land holding left the Irish people vulnerable to poverty and food insecurity, and the practice of eviction made it difficult for them to find food and support during the crisis. The British government's response to the famine was heavily influenced by the interests of the wealthy landlords, making it difficult for the Irish people to access aid and support. The Irish Famine serves as a reminder of the importance of fair and equitable land holding systems in ensuring food security and preventing disaster and crisis (OpenAI 2023).

This text was then analysed by the class. Students interestingly first critiqued the writing and felt that while it was reasonable it was more in line with the type of writing done in secondary school rather than third level. Due to the focus on writing skills in the first semester of their Critical Skills course this does not come as a surprise to the author. The students by the second semester were well able to identify that the writing did not have the structure or analysis that is expected in higher education. However, there was little criticism of the content. Given that these students lacked knowledge of the topic (as they are first years and from a range of disciplines) it was necessary to guide them through the information and interpretation present in the text. This provided an opportunity to demonstrate a central feature of LLMs, they are prone to producing plausible sounding but inaccurate information (as alluded to earlier they produce statistically viable text and are not drawing on real information).

While the first two paragraphs demonstrate a poor grasp of the nuances of land holding leading up to the famine, the third is particularly problematic. One of the tenets of British policy was that 'Irish property should pay for Irish poverty' and landlord interests railed against the introduction of an Irish Poor Law (passed in 1838). In addition, the repeal of the Corn Laws (tariffs introduced in 1815 to protect agricultural prices (and hence landowners) following the ending of the Napoleonic Wars) in 1846 and

1849 in response to the famine was strongly opposed by landlords as it reduced agricultural prices (Bartlett 2010, 272; Cannon and Crowcroft 2015). Hence, the claim by ChatGPT that the government's response was 'heavily influenced by the interests of the wealthy landlords' is not factually correct (Donnelly 1989; Gray 2014). Also, the 'Protection of Tenants Act of 1847' does not exist.

To further emphasise the difficulties with 'hallucinations' (i.e. the false information produced by LLMs), ChatGPT was asked to produce a reading list on this topic. In response it produced 10 books which students were asked in groups to investigate. What they discovered was it had produced a list of sources of which three existed, one did not, and a further six contain some correct information (mainly an author that has written on the Irish famine with a non-existent title) (see appendix 4). It should be noted that such lists on more obscure topics and the production of academic references are even more problematic.

This analysis established two central issues with the integration of ChatGPT into student work. Firstly, without adequate understanding and experience of academic writing, students would have difficulty critiquing the text produced by LLMs. Secondly, for those lacking disciplinary knowledge, analytical and critical thinking, research skills and the ability to synthesise scholarly sources to produce original arguments, LLMs are potentially detrimental to students in researching essays. Thus, in reference to the literacies identified by Southworth et al. (2023), students knew and understood what generative AI is and how it operates. Following this instructor-led engagement, students then had the opportunity to engage with LLMs first hand.

The remainder of the class provided an opportunity for students to apply further critical literacies including using and applying generative AI, and critically evaluating AI (Concannon et al. 2023; Southworth et al. 2023). To begin with students used ChatGPT to work on their essay topic from the first semester of the course. This allowed them to operate the LLM, while also ensuring they had the knowledge and understanding to critique the content produced, a central element in the successful use of the technology (Sabzalieva and Valentini 2023). Given the small size of the class groups, the course instructor was available during this time to support students in their work. Following the generation of essays, the class finished with a student lead discussion of the affordances and limitations of the technology. The consensus was the technology could prove useful for routine writing tasks that I am going to term 'bullshit writing', i.e. writing that is highly standardised (e.g. emails). For academic writing, potential uses were mooted, including generating ideas, helping with writers block, and as a writing assistant; however, the students felt it would not be possible to produce good academic assignments relying primarily on the technology.³

Discussion and conclusion

While ChatGPT is new, educational technologies have a long history of promises to radically change the content, delivery and purpose of education, as well as raising concerns about the integrity of academic assessment (Blikstein and Blikstein 2021). If Science and Technology Studies scholars and Historians of Technology have taught us anything it is that technologies rarely develop in the manner in which we think. Rather technology use is an interplay of technological, social, cultural, political and economic factors (Dick 2020; Hekkert et al.

2007; Oudshoorn and Pinch 2003). We must thus be careful not to engage in kneejerk reactions to emerging technologies. For example, during the recent Covid-19 pandemic, claims abound about the need to equip students and workers with the skills necessary to negotiate a future of online work (Bayerlein et al. 2021; Lund et al. 2021). This discourse has subsequently diminished primarily due to the realisation of a significant number of problems that have been caused by the lockdown, combined with a substantial reversal of work-from-home policies. Added to this inability to accurately predict the future, we must also be careful to avoid the boosterism that has accompanied the launch of ChatGPT. Many commentators are predicting a productivity revolution as these LLMs are integrated into work with claims that far exceed the technology's capability.

Thus, central to our assessment of potential educational technology is a realistic understanding of how it operates in the here and now. Yes, LLMs will develop but to predict in what way and how this will be utilised is to engage in a form of futurism that has rarely been successful. Thus, this article argues that developing transferable critical AI literacies will equip students in reacting to future technological developments. To do so, educators will need to be able to understand the technology and its operation. Given its scores on standardised exams, there is no denying that it can do some things extremely well. As such, educators must not underestimate the 'disruptive' impact it is and will have. However, it is not human and teachers must appraise it to understand the higher thinking skills that will become increasingly important and how we teach and assess these. With this knowledge they can then aid students in their exploration of the potential uses of LLMs in their own work practice, while also providing them with the tools necessary to properly critique its limitations. It is hoped that the discussion above can aid in beginning this discourse in regard to instructor understanding and how the topic of LLMs might be broached in a classroom setting. In doing so, we must be critical: yes, LLMs, in their current configuration, can be useful writing assistants but they cannot do the important tasks of researching, analysing, and synthesising that are at the core of academic work.

While improvements have been noted in newer models, such as GPT-4 which has made significant strides in standardised tests, the architecture of current generative AI means that hallucinations will remain (Eloundou et al. 2023). Bender et al. (2021) have warned of the dangers of larger and larger LLMs which include 'stereotyping, denigration, increases in extremist ideology' and other such issues as 'humans encounter seemingly coherent LM [sic.] output' (Bender et al. 2021, 619). It is thus important that educators continue to engage with these technologies as they evolve to ensure that their understanding of the affordances and limitations of LLMs remains current (NAIN 2023). The case study provided above is of course time sensitive and reflects the technology's capability as it existed at that moment in time. As such, educators must be adaptive to technological developments and adjust learning activities to reflect these, while still teaching critical AI literacies. The appearance of peer-reviewed articles written with AI assistance raises questions as to the potential of generative AI in the writing process – it should be noted that such use obviously raises research integrity and ethics concerns, including the potential privacy concerns (Stahl and Eke 2024). However, notwithstanding these issues, such expert use relies on a strong understanding of the subject matter. For many students, particularly undergraduates, such expertise is lacking. This in turn can have detrimental effects on their studies. The development of critical literacies is thus essential if students

are to navigate the emergence of generative AI. They must be equipped with the relevant literacies to understand how AI works, how to operate it and how to critique it. Drawing on the ARCL information literacies framework (ARCL 2015) as a basis, it is important to understand that while such knowledge is essential, students must develop dispositions toward AI that can be applied to these evolving technologies in various contexts.

Despite the limitations of generative AI, students will still be drawn to it as they engage with academic assessment. A number of actions can mitigate against such use. Firstly, as this article has outlined, student understanding of the limitations of generative AI and its potential harm to their learning is an important element in discouraging its use. Likewise, a strong institutional AI policy (that of course allows instructors to introduce and assign AI assisted tasks when appropriate) will also remove ambiguity around the ethicality of AI use and such guidance is essential in reducing plagiarism (Cronan, Mullins, and Douglas 2018; NAIN 2023). Secondly, generative AI frames writing as a product not a process. Consequently, assessment of, for and as learning must form a central component of assessment practices (National Forum for the Enhancement of Teaching and Learning 2016). Such an approach focusing on research and writing processes will mean moving away from high-stakes summative assessments, to low-stakes and formative assessment practices that break the writing and research process into stages (NAIN 2023). Such an assessment approach has a number of advantages in enhancing academic integrity, including built-in time management, reduced pressure from high-stakes assessment, and a body of work to draw on in completion of final assessments (which is also useful to instructors in assessing a potential case of plagiarism). In essence, completing the assessment as intended becomes the path of least resistance. The introduction of peer review and reflective exercises can also reduce pressures on graders, while enhancing metacognition. The tasks and learning outcomes we assign are also central to the potential use of ChatGPT and other LLMs. Thus, assignments that ask students to synthesise research, examine contrasting and evolving academic discourses, and utilise this to produce original insights are central to such assessment practices. This is essential for protecting academic integrity but also in ensuring that we assess what is truly important to the mission of higher education: creating critically aware, analytical thinkers who have the power to shape the world around them.

Notes

1. Readers should note that GPT-3 was released in 2020. In November 2022 OpenAI released ChatGPT which utilises GPT-3.5, a updated version of GPT-3. In March 2023 GPT-4 was released, the subscription version of ChatGPT utilises this model. All examples in this article are produced by the free version of ChatGPT which utilises GPT-3.5.
2. The author would like to point out that no AI was harmed (or used) in the production of this article.
3. It should be noted that the release of GPT-4 has reduced the level of hallucinations and as these technologies develop this will impact the ways in which students utilise them.

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Appendices

Appendix 1 Write a business plan for an exploration of mars



Executive Summary:

The purpose of this business plan is to outline the feasibility of launching a Mars exploration mission. The objective of the mission is to study the geology, climate, and potential for human habitation on Mars. The ultimate goal is to establish a permanent human settlement on Mars, which would provide numerous benefits to both humanity and the advancement of scientific research.

Market Analysis:

The market for space exploration and the study of Mars is growing rapidly. With the increasing interest in space exploration and the development of new technologies, the demand for Mars exploration missions is expected to continue to rise. This demand is driven by both government agencies and private organisations, which are investing in space exploration to advance scientific research and to secure a future for humanity beyond Earth.

Product and Service Offerings:

The Mars exploration mission will offer a wide range of services and products, including scientific research, the development of new technologies, and the establishment of a permanent human settlement on Mars. The mission will also provide a platform for scientific experiments and the study of the Martian environment, which will have a significant impact on our understanding of the planet and the potential for human habitation.

Marketing and Sales:

The Mars exploration mission will be marketed to government agencies, private organisations, and the general public. The mission will also be marketed to universities and research institutions to attract scientific talent and funding for the mission. Additionally, the mission will generate revenue through the sale of products and services related to the exploration and study of Mars.

Operations:

The Mars exploration mission will be operated by a team of experienced scientists and engineers, with support from government agencies and private organisations. The mission will be supported by a network of international partners, including universities and research institutions, to ensure the success of the mission and the advancement of scientific research.

Financial Projections:

The estimated cost of the Mars exploration mission is approximately \$10 billion. This includes the development and launch of the spacecraft, the support and operation of the mission, and the establishment of a permanent human settlement on Mars. The mission will generate significant revenue through the sale of products and services related to the exploration and study of Mars, which will offset the cost of the mission.

Conclusion:

The Mars exploration mission is a bold and innovative project that will have a significant impact on the advancement of scientific research and the future of humanity. The mission offers numerous opportunities for growth and revenue, and the demand for space exploration and the study of Mars is expected to continue to rise. The success of the mission will be driven by a team of experienced scientists and engineers, with support from government agencies, private organisations, and a network of international partners. (OpenAI, 2023).

Appendix 2 Now write this in the style of a pirate

Ahoy, mateys! 'Tis time to set sail on a grand adventure to the red planet, Mars! We be embarkin' on a mission of exploration, with a goal of studyin' the geology, climate, and potential for pillagin' and colonizin' the new world.

The market be ripe for this journey, with many a government agency and private organisation investin' in space exploration and the search for new treasures. Our offerings be vast, from scientific research to new technologies and even the establishin' of a permanent pirate settlement on Mars.

We'll be marketin' our mission to any and all who be interested, from government agencies to universities and research institutions, and even the general public. Our product and service offerings will bring in riches beyond measure, and we'll be sure to make the most of every opportunity for profit.

Our operations be run by a crew of experienced scientists and engineers, with support from government agencies and private organisations, as well as a network of international pirate partners. The cost of this voyage be estimated at ten million pieces of eight, but fear not, mateys, as the sale of our products and services related to the exploration and pillage of Mars will more than make up for it.

So hoist the Jolly Roger and set sail on this grand adventure, me hearties! The future of piracy and the advancement of science be in our grasp, and the treasure to be had is limitless! Yarr!

Appendix 3 write a 500 word essay on the causes of the irish famine

The Irish Famine, also known as the Great Hunger, was a catastrophic event in Irish history that took place between 1845 and 1852. It was a period of extreme poverty, suffering, and death caused by the failure of the potato crop, the main staple food of the Irish people. The famine had a profound impact on Ireland and its people, causing mass emigration and reducing the population of the country by approximately 25%.

The primary cause of the Irish Famine was the failure of the potato crop. The potato was the staple food of the Irish people and was relied upon by the vast majority of the population for sustenance. In 1845, a fungal disease known as late blight spread rapidly across the country, destroying the potato crop and leaving the Irish people without their main source of food. The failure of the potato crop was exacerbated by the fact that the Irish were heavily dependent on a single variety of potato, making them vulnerable to disease and crop failure.

Another cause of the famine was the poor economic and social conditions in Ireland at the time. The country was a part of the United Kingdom, but it was largely controlled by the British government, which had a limited understanding of the challenges facing Ireland. The country was primarily an agricultural society, with a large portion of the population living in poverty and relying on the potato crop for their livelihood. Furthermore, the British government imposed high tariffs on Irish goods, making it difficult for the country to export its products and earn revenue.

In addition, the British government's response to the famine was inadequate and slow. Instead of providing immediate assistance to the Irish people, the government focused on laissez-faire policies, which encouraged people to rely on the market for food and support. The government also believed that the famine was a natural disaster and that it was not their responsibility to provide aid. This attitude was reflected in the lack of government aid and the slow response to the crisis, which allowed the famine to continue for several years and cause widespread suffering and death.

Furthermore, the Irish people were also facing discrimination and prejudice from the British government and society. The Irish were seen as inferior and were treated as second-class citizens by the British. This discrimination extended to the response to the famine, where the Irish were often denied access to food and aid while it was readily available to other parts of the United Kingdom.

Another factor that contributed to the severity of the famine was the lack of diversification in the Irish economy. The country was heavily dependent on agriculture, with the vast majority of the population relying on the potato crop for their livelihood. This lack of diversification made the country vulnerable to economic shocks, such as the failure of the potato crop, and limited the options available to the Irish people in the face of the crisis.

In conclusion, the Irish Famine was a catastrophic event caused by a combination of factors, including the failure of the potato crop, poor economic and social conditions, inadequate government response, discrimination, and lack of diversification in the Irish economy. The famine had a profound impact on Ireland and its people, causing mass emigration and reducing the population of the country by approximately 25%. It is a dark chapter in Irish history that serves as a reminder of the importance of government responsibility and action in the face of disaster and crisis.

Appendix 4 produce a reading list for this topic

1. "The Great Irish Famine: Impact, Ideology and Rebellion" by Christine Kinealy
2. "The Irish Famine: A Documentary" edited by Christine Kinealy and Gerard Moran
3. "The Famine Plot: England's Role in Ireland's Greatest Tragedy" by Tim Pat Coogan
4. "The Famine in Ireland" by Cormac O Grada
5. "The Making of the Irish Poor Law, 1815-1840" by Laurence M. Geary
6. "The People's Food: A History of the Irish Famine" by Terry Breverton
7. "The Irish Famine: An Illustrated History" by John Burke
8. "A Painful Business: The British Government and the Famine in Ireland 1845-1852" by Ciarán Ó Murchadha
9. "The Irish Famine: A Global History" by Christine Kinealy